

Topic 14: Mechanical Oscillations, Damping & Pure Resonance

Professor Leonard Master Series · Mechanical Vibrations

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Bouncing Mass on a Spring

"Picture a mass m hanging from a spring with stiffness k , inside a shock absorber with damping β , pushed by an external force $F(t)$."

Newton's Second Law ($F_{\text{net}} = ma$):

$$mx'' = -kx - \beta x' + F(t) \implies mx'' + \beta x' + kx = F(t)$$

2. Curriculum Reference Diagrams

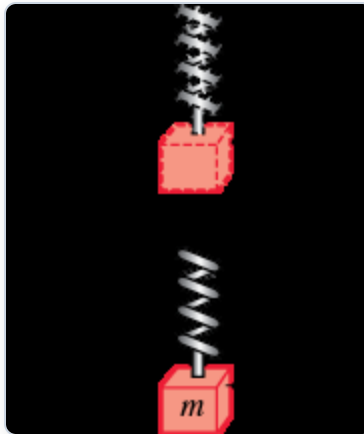


Figure 14.1: Mass-spring coordinate reference — from Zill 7th Ed. Chapter 3 (Fig. 3.8.2).

$$x(t) = \frac{\sqrt{3}}{6} \cos(8t - (-0.245)) \quad \text{or} \quad x(t) = \frac{\sqrt{3}}{6} \cos(8t + 0.245). \quad (9') \equiv$$

■ Graphical Interpretation

FIGURE 3.8.4(a) illustrates the mass in Example 2 going through approximately two complete cycles of motion. Reading left to right, the first five positions marked with black dots in the figure correspond, respectively,

- to the initial position ($x = \frac{2}{3}$) of the mass below the equilibrium position,
- to the mass passing through the equilibrium position ($x = 0$) for the first time heading upward,

Figure 14.2: Damping behavior: Underdamped envelope decay vs Overdamped slow return — from Zill 7th Ed. Chapter 3 (Fig. 3.8.4).

3. The Three Damping Regimes

Discriminant: $\Delta = \beta^2 - 4mk$.

1. **Overdamped** ($\beta^2 > 4mk$): Heavy damping (like moving through honey). Returns to rest with no oscillations.
2. **Critically Damped** ($\beta^2 = 4mk$): The sweet spot for car shock absorbers. Returns to equilibrium as fast as physically possible without overshooting.
3. **Underdamped** ($\beta^2 < 4mk$): Bounces back and forth with decaying amplitude:

$$x(t) = Ae^{-\frac{\beta}{2m}t} \cos(\omega_d t - \phi)$$

4. Pure Resonance: When Buildings Collapse

If there is NO damping ($\beta = 0$) and the driving frequency equals the natural frequency ($\omega = \omega_0 = \sqrt{k/m}$):

$$x'' + \omega_0^2 x = F_0 \cos(\omega_0 t)$$

Because $\cos(\omega_0 t)$ is in y_c , undetermined coefficients requires multiplying by t :

$$x_p(t) = \frac{F_0}{2\omega_0} t \sin(\omega_0 t)$$

The amplitude grows linearly with time $t \rightarrow \infty$ without bound! This is what collapsed the Tacoma Narrows Bridge!